



ABBOTSLEIGH

August 2002
TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using blue or black pen.
- Board-approved calculators may be used.
- A table of standard integrals is provided.
- All necessary working should be shown in every question.

Total marks (84)

- Attempt Questions 1-7.
- All questions are of equal value.

Total marks – 84
Attempt Questions 1-7
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

QUESTION 1 (12 Marks) Use a SEPARATE writing booklet.

Marks

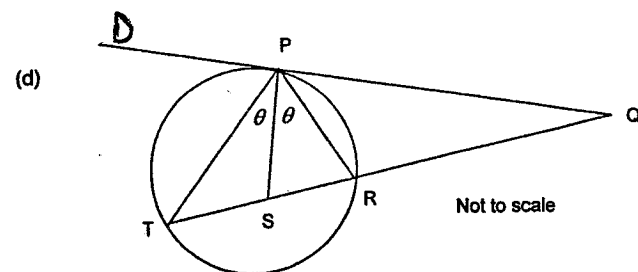
- (a) Differentiate
- (i) $\log_r(3x^2 + 2)$ (1)
- (ii) $(1 + x^2)\tan^{-1}x$. (2)
- (b) Solve the inequality $\frac{2x}{x-2} \leq 3$ (3)
- (c) Evaluate exactly $\int_1^{\sqrt{5}} \frac{dt}{\sqrt{4-t^2}}$ (2)
- (d) Using the substitution $u = 4 - x$ evaluate $\int_3^4 x\sqrt{4-x} \, dx$. (4)

QUESTION 2 (12 Marks) Use a SEPARATE writing booklet.

(a) Evaluate $\int_0^{\pi} \cos^2 x \, dx$ (3)

(b) Show that $x+1$ is a factor of $x^3 - 4x^2 + x + 6$.
Hence or otherwise, factorise $x^3 - 4x^2 + x + 6$ fully. (3)

(c) The equation $x^3 + 2x - 8 = 0$ has a root close to $x = 1.6$. Use one application of Newton's method to find a better approximation to the root. (Give your answer to 2 decimal places). (3)



In the diagram the vertices of triangle PTR lie on a circle. The tangent at P meets the secant TR produced at Q . The bisector of $\angle TPR$ meets TR at S .

Copy the diagram into your booklet.
Prove that $PQ = SQ$.

(3)

QUESTION 3 (12 Marks) Use a SEPARATE writing booklet.

Marks

(a) (i) State the domain and range of $y = 3\cos^{-1} 2x$ (2)

(ii) Find the value of y if $x = \frac{1}{4}$ (1)

(iii) Sketch the graph of $y = 3\cos^{-1} 2x$. (1)

(b) Let α, β, γ be the roots of the polynomial $3x^3 - 12x^2 - 8 = 0$.
Evaluate $\alpha\beta\gamma$. (2)

(c) If $\sin A = \frac{2}{3}$ and $\frac{\pi}{2} < A < \pi$, find the exact value of $\sin 2A$ (2)

(d) The acceleration of a particle x metres from 0 at time t seconds is given by

$$\frac{d^2x}{dt^2} = -e^{-2x}$$

If the velocity is 1 metre per second when $x = 0$, find the exact velocity when $x = 4$ metres.

(4)

QUESTION 4 (12 Marks) Use a SEPARATE writing booklet.

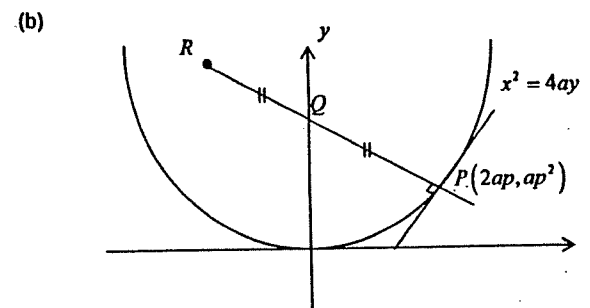
Marks

- (a) Solve $\sqrt{3} \cos x + \sin x = 1$ for $0 \leq x \leq 2\pi$. (4)
- (b) (i) Explain why the function $f(x) = \sqrt{x-2}$ has an inverse function $f^{-1}(x)$. (1)
- (ii) Write down the equation of the inverse function $f^{-1}(x)$ and sketch both $y = f(x)$ and $y = f^{-1}(x)$ on the same set of axes. (3)
- (c) (i) Express $\sin A$ and $\cos A$ in terms of t where $t = \tan \frac{A}{2}$. (1)
- (ii) Hence or otherwise prove that $\frac{\sin 2A}{1 + \cos 2A} = \tan A$. (3)

QUESTION 5 (12 Marks) Use a SEPARATE writing booklet.

Marks

- (a) Given that $f(x) = \frac{x}{4-x^2}$
- (i) Determine whether $f(x)$ is odd, even or neither. (1)
- (ii) Show that $f(x)$ has no stationary points. (3)
- (iii) Find any horizontal or vertical asymptotes. (2)



The normal at $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ cuts the y -axis at Q and is produced to a point R such that $PQ = QR$.

- (i) Given that the equation of the normal at P is $x + py = 2ap + ap^3$, find the coordinates of Q . (1)
- (ii) Show that R has coordinates $(-2ap, ap^2 + 4a)$. (2)
- (iii) Show that the locus of R is a parabola and state its vertex. (3)

QUESTION 6 (12 Marks) Use a SEPARATE writing booklet.

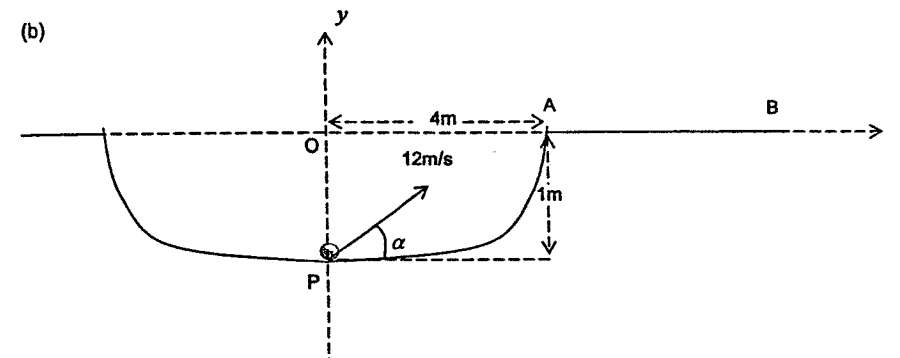
Marks

- (a) A point moves along the curve $y = \frac{1}{x}$ such that the x coordinate is changing at the rate of 2 units per second. At what rate is the y coordinate decreasing when $x = 5$? (3)
- (b) Molten metal at a temperature of 1400 °C is poured into moulds to form machine parts. After 15 minutes the metal has cooled to 995°C. If the temperature after t minutes is T °C, and if the temperature of the surroundings is 35°C, then the rate of cooling is approximately given by $\frac{dT}{dt} = -k(T - 35)$ where k is a positive constant. (405)
- (i) Show that a solution of this equation is $T = 35 + Ae^{-kt}$ where A is a constant. (1)
- (ii) Find the values of A and k . (3)
- (iii) The metal can be taken out of the moulds when its temperature has dropped to 200°C. How long after the metal has been poured will this temperature be reached? (2)
- (c) Prove by mathematical induction that $2^{3n} - 3^n$ is divisible by 5 for all positive integers n . (3)

QUESTION 7 (12 Marks) Use a SEPARATE writing booklet.

Marks

- (a) Find $\lim_{x \rightarrow 0} \frac{3x}{\tan 4x}$ (1)



A golf ball is lying at point P, at the middle of the bottom of a sand bunker which is surrounded by level ground. The point A is at the edge of the bunker 4 m from O and AB lies on level ground. The initial velocity is 12 m/s and P is 1 m below O.

- (i) Using $g = -10 \text{ m/s}^2$, show that the golf ball's trajectory at time t seconds after being hit may be defined by the equations:
- $$x = (12 \cos \alpha)t \quad \text{and} \quad y = -5t^2 + (12 \sin \alpha)t - 1$$
- where x and y are the horizontal and vertical displacements, in metres, of the ball from the origin O shown in the diagram, and α is the angle of projection. (3)
- (ii) Given $\alpha = 30^\circ$, how far from A will the ball land? (3)
- (iii) Find the range of values of α , to the nearest degree, at which the ball must be hit so that it will land to the right of A. (4)

END OF PAPER

1) a) (i) $\frac{6x}{3x^2+2}$
 (ii) $(1+x^2) \times \frac{1}{1+x^2} + \tan^{-1} x \times 2x$
 $= 1 + 2x \tan^{-1} x$

b) $\frac{2x}{x-2} \leq 3 \quad x \neq 2$
 $(x-2) \times \frac{2x}{x-2} \leq 3(x-2)$

$2(x-2)^2 - 2x(x-2) \geq 0$
 $(x-2)(2(x-2)-2x) \geq 0$
 $(x-2)(x-6) \geq 0$
 $x < 2 \text{ or } x \geq 6$

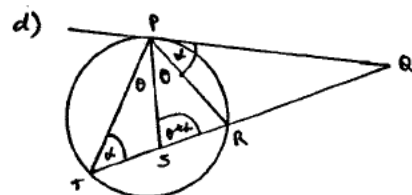


c) $\int_1^{\sqrt{3}} \frac{dt}{\sqrt{4-t^2}} = \left[\sin^{-1}\left(\frac{t}{2}\right) \right]_1^{\sqrt{3}}$
 $= \sin^{-1}\frac{\sqrt{3}}{2} - \sin^{-1}\frac{1}{2}$
 $= \frac{\pi}{3} - \frac{\pi}{6}$
 $= \frac{\pi}{6}$

d) $\int_3^4 x \sqrt{4-x} dx$
 $u = 4-x$
 $du = -dx$
 $x = 3 \Rightarrow u = 1$
 $x = 4 \Rightarrow u = 0$
 $= \int_1^0 -(4-u) \sqrt{u} du$
 $= \int_0^1 (4-u) \sqrt{u} du$
 $= \left[\frac{2 \times 4 u^{3/2}}{3} - \frac{2u^{5/2}}{5} \right]_0^1$
 $= \left(\frac{8 \times 1^{3/2}}{3} - \frac{2 \times 1^{5/2}}{5} \right) - (0-0)$
 $= \frac{8}{3} - \frac{2}{5} = \frac{34}{15} \text{ or } 2\frac{4}{15}$

b) $f(x) = x^3 - 4x^2 + x + 6$
 $f(-1) = (-1)^3 - 4(-1)^2 - 1 + 6 = -1 - 4 - 1 + 6 = 0$
 $\therefore x+1$ is a factor of $f(x)$
 $f(x) = (x+1)(x-2)(x-3)$

c) $f(x) = x^3 + 2x - 8$
 $f'(x) = 3x^2 + 2$
 $f(1.6) = (1.6)^3 + 2(1.6) - 8 = -0.704$
 $f'(1.6) = 3(1.6)^2 + 2 = 9.68$
 $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 1.6 - \frac{-0.704}{9.68} = 1.6727273$
 $= 1.67$

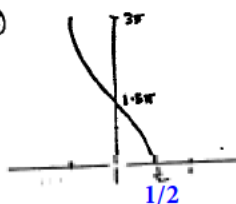


$\alpha = \angle QPR = \angle PTR$ (angle between tangent & chord equals angle in alternate segment)
 $\angle PSR = \alpha + \theta$ (exterior angle of Δ = sum of 2 interior opp angles)
 $\angle QPS = \alpha + \theta$ (by addition)
 $\therefore PQ = SQ$ (sides opp equal angles)

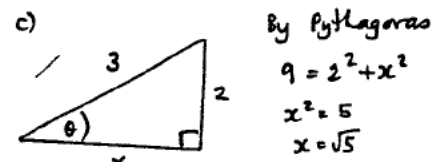
8) a) (i) Domain of $\cos^{-1} x$ $-1 \leq x \leq 1$
 Range of $\cos^{-1} x$ $0 \leq y < \pi$

$y = \cos^{-1} x$
 $\frac{y}{3} = \cos^{-1} 2x$
 Domain $-1 \leq 2x \leq 1$
 $\therefore -\frac{1}{2} \leq x \leq \frac{1}{2}$
 Range $0 \leq \frac{y}{3} \leq \pi$
 $0 \leq y \leq 3\pi$

(ii) $y = 3 \cos^{-1}\left(\frac{2}{3}\right)$
 $= 3 \cos^{-1}\left(\frac{1}{3}\right)$
 $= 3 \times \frac{\pi}{3}$
 $= \pi$



b) $\alpha\beta\gamma = \text{product of roots}$
 $3x^3 - 12x^2 + 0x - 8 = 0$
 $\alpha\beta\gamma = -\frac{d}{a}$
 $\alpha\beta\gamma = \frac{8}{3}$

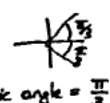


$\frac{\pi}{2} < A < \pi$ 2nd quadrant
 $\sin A = \frac{2}{3}$ $\cos A = -\frac{\sqrt{5}}{3}$
 $\sin 2A = 2 \sin A \cos A$
 $= 2 \times \frac{2}{3} \times -\frac{\sqrt{5}}{3}$
 $= -\frac{4\sqrt{5}}{9}$

d) $\frac{d^2x}{dt^2} = -e^{-2x}$
 $\text{acc} = \frac{dv}{dx} \left(\frac{1}{2} v^2 \right)$
 $\frac{dv}{dx} \left(\frac{1}{2} v^2 \right) = -e^{-2x}$
 $\frac{1}{2} v^2 = \frac{+e^{-2x}}{2} + C$
 When $x=0$ $v=1$
 $\frac{1}{2} = \frac{e^0}{2} + C$
 $\therefore C = 0$
 $\frac{1}{2} v^2 = \frac{e^{-2x}}{2}$
 $v^2 = e^{-2x}$
 $v = \pm \sqrt{e^{-2x}}$
 $v = e^{-x}$ (take +ve as $v=1$ when $x=0$)
 When $x=4$ $v = e^{-4}$
 $v = \frac{1}{e^4}$ metres per second

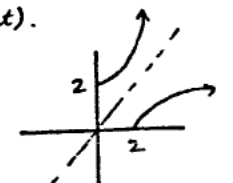
4) a) $\sqrt{3} \cos x + \sin x = 1 \quad 0 \leq x \leq 2\pi$
 Let $\sqrt{3} \cos x + \sin x = A \cos(x-\alpha)$
 $= A \cos x \cos \alpha + A \sin x \sin \alpha$
 $A \cos \alpha = \sqrt{3}$ (1)
 $A \sin \alpha = 1$ (2)
 $A^2 (\sin^2 \alpha + \cos^2 \alpha) = (\sqrt{3})^2 + 1^2$
 $A^2 = 4$
 $A = \pm 2$ take positive $A = 2$
 $(2) \div (1) \quad \tan \alpha = \frac{1}{\sqrt{3}}$ First quadrant as $\sin \alpha > 0$
 $\alpha = \frac{\pi}{6}$
 $2 \cos(x - \frac{\pi}{6}) = 1$
 $\cos(x - \frac{\pi}{6}) = \frac{1}{2}$
 $0 \leq x \leq 2\pi$
 $0 - \frac{\pi}{6} \leq x - \frac{\pi}{6} \leq 2\pi - \frac{\pi}{6}$
 $-\frac{\pi}{6} \leq x - \frac{\pi}{6} \leq \frac{11\pi}{6}$

$x - \frac{\pi}{6} = \frac{\pi}{3} \text{ or } 2\pi - \frac{\pi}{3}$
 $x - \frac{\pi}{6} = \frac{\pi}{3}, \frac{5\pi}{3}$
 $x = \frac{\pi}{3} + \frac{\pi}{6}, \frac{5\pi}{3} + \frac{\pi}{6}$
 $= \frac{3\pi}{6}, \frac{11\pi}{6}$
 $x = \frac{\pi}{2}, \frac{11\pi}{6}$



b) (i) $y = \sqrt{x-2}$ has an inverse function because it is a one-to-one function. (horizontal line test).

(ii) $x = \sqrt{y-2}$
 $x^2 = y-2$
 $y = x^2 + 2$
 $f'(x) = 2x$ for $x > 0$
 Inverse function $f^{-1}(x)$ is restricted to $x > 0$ since it is only half the parabola.
 c) if $t = \tan \frac{A}{2}$ $\sin A = \frac{2t}{1+t^2}$ $\cos A = \frac{1-t^2}{1+t^2}$



(ii) $\frac{\sin 2A}{1 + \cos 2A} = \tan A$
 Let $t = \tan A$ from above
 $\sin 2A = \frac{2t}{1+t^2}$ $\cos 2A = \frac{1-t^2}{1+t^2}$
 LHS $= \frac{2t}{1+t^2} \div \left(\frac{1+t^2}{1+t^2} \right)$
 $= \frac{2t}{1+t^2} \times \frac{1+t^2}{1+t^2}$
 $= \frac{2t}{1+t^2} \times \frac{1+t^2}{2}$
 $= t$
 $= \tan A$
 $= \text{RHS}$

2) a) $\int_0^{\pi} \cos^2 x dx = \int_0^{\pi} \frac{1 + \cos 2x}{2} dx$
 $= \frac{1}{2} \left[x + \frac{\sin 2x}{2} \right]_0^{\pi}$
 $= \frac{1}{2} \left[\pi + \frac{\sin 2\pi}{2} \right] - \left(0 + \frac{\sin 0}{2} \right)$
 $= \frac{1}{2} \left[\pi + 0 \right] - 0$
 $= \frac{\pi}{2}$

5) a) $f(x) = \frac{x}{4-x^2}$
 (i) $f(-x) = \frac{-x}{4-(-x)^2} = \frac{-x}{4-x^2} = -f(x)$
 $\therefore$ odd function
 (ii) $f'(x) = \frac{(4-x^2) \cdot 1 - x(-2x)}{(4-x^2)^2}$
 $= \frac{4-x^2+2x^2}{(4-x^2)^2}$
 $= \frac{4+x^2}{(4-x^2)^2}$

$$4+x^2 \neq 0 \text{ since } 4+x^2 > 0 \text{ for all values}$$

of x (since x^2 is always positive)

$\therefore$ since $f'(x) \neq 0$ there are no stat pts.

$$\begin{aligned} \text{As } x \rightarrow \infty \quad \lim_{x \rightarrow \infty} \frac{x}{4-x^2} \\ = \lim_{x \rightarrow \infty} \frac{\frac{x}{x^2}}{\frac{4-x^2}{x^2}} = \frac{\frac{1}{x}}{\frac{4}{x^2}-1} \\ = \frac{0}{-1} = 0 \end{aligned}$$

$y=0$ is an asymptote (horizontal)
Denominator $4-x^2 \neq 0$

$f(2-x)(2+x)=0$
 $x=2, x=-2$ are asymptotes (vertical)

$$\begin{aligned} \text{b) (i) } x+py &= 2ap+ap^3 \\ \text{sub in } x=0 \quad py &= 2ap+ap^3 \\ y &= 2a+ap^2 \end{aligned}$$

$$Q(0, 2a+ap^2) \quad P(2ap, ap^2)$$

Q is midpt so

$$\begin{aligned} \frac{x+2ap}{2} = 0 \quad \text{and} \quad \frac{y+ap^2}{2} = 2a+ap^2 \\ x+2ap=0 \quad y+ap^2=4a+2ap^2 \\ x=-2ap \quad y=4a+ap^2 \end{aligned}$$

$\therefore R$ is $(-2ap, 4a+ap^2)$

$$\begin{aligned} \text{(ii) } x=-2ap \quad y=4a+ap^2 \\ p = \frac{-x}{2a} \text{ sub into } y \\ y = 4a + a\left(\frac{-x}{2a}\right)^2 = 4a + \frac{x^2}{4a^2} \\ y = \frac{16a^3 + x^2}{4a^2} \\ y = \frac{16a^2 + x^2}{4a} \quad 4ay = x^2 + 16a^2 \\ x^2 = 4a(y-4a) \end{aligned}$$

parabola vertex $(0, 4a)$.

$$\text{b) (a) } \frac{dx}{dt} = 2$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{dx}{dt} \times \frac{dy}{dx} \\ &= 2x - \frac{1}{x^2} \end{aligned}$$

When $x=5$

$$\begin{aligned} \frac{dy}{dx} &= 2 \times 5 - \frac{1}{25} \\ &= \frac{-2}{25} \end{aligned}$$

$$\begin{aligned} y &= \frac{1}{x} = x^{-1} \\ \frac{dy}{dx} &= -x^{-2} \\ &= -\frac{1}{x^2} \end{aligned}$$

$$\begin{aligned} \text{b) (i) } T &= 35 + Ae^{-kt} \quad (1) \\ \frac{dT}{dt} &= -kAe^{-kt} \\ &= -k(T-35) \text{ from (1)} \\ T &= 35 + Ae^{-kt} \text{ is a soln to } \frac{dT}{dt} = -k(T-35) \end{aligned}$$

$$\begin{aligned} \text{(ii) } t=0 \quad T &= 1400 \\ t=15 \quad T &= 995 \\ \text{When } t=0 \quad 1400 &= 35 + Ae^0 \\ A &= 1365 \\ t=15 \quad 995 &= 35 + 1365e^{-15k} \\ \frac{960}{1365} &= e^{-15k} \end{aligned}$$

$$\begin{aligned} \log_e \left(\frac{960}{1365} \right) &= -15k \\ k &= \frac{-1}{15} \log_e \left(\frac{960}{1365} \right) \\ k &= 0.027465094 \end{aligned}$$

$$\begin{aligned} \text{(iii) } T &= 200 \quad t = ? \quad -0.027465t \\ 200 &= 35 + 1365e^{-0.027465t} \\ \frac{165}{1365} &= e^{-0.027465t} \\ t &= \frac{\ln \left(\frac{165}{1365} \right)}{0.027465} = 9.04712 \\ \text{It will take } &9.04712 \text{ minutes} \end{aligned}$$

$$\begin{aligned} \text{c) } 2^{3n} - 3^n &\text{ is divisible by 5} \\ \text{Prove true for } n=1 \\ 2^3 - 3^1 &= 8 - 3 = 5 \\ \therefore \text{ True for } n=1 \\ \text{Let it be true for } n=k \\ 2^{3k} - 3^k &= 5m \text{ where } m \text{ is a positive integer} \\ \therefore 2^{3k} &= 5m + 3^k \\ \text{Prove true for } n=k+1 \\ 2^{3(k+1)} - 3^{k+1} &= 2^3 \times 2^{3k} - 3^{k+1} \\ &= (5m + 3^k) \times 8 - 3^k \times 3 \\ &= 40m + 8 \times 3^k - 3^k \times 3 \\ &= 40m + 5 \times 3^k \\ &= 5(8m + 3^k) \text{ which is} \end{aligned}$$

divisible by 5 if m is a positive integer.
If it is true for $n=k$ we have proven it true for $n=k+1$. Since it is true for $n=1$ then it is true for $n=1+1=2$ and so on for all positive integral n .

$$\begin{aligned} \text{a) } \lim_{x \rightarrow 0} \frac{3x}{\tan 4x} &= \frac{3}{4} \lim_{x \rightarrow 0} \frac{4x}{\tan 4x} \\ &= \frac{3}{4} \times 1 \\ &= \frac{3}{4} \end{aligned}$$

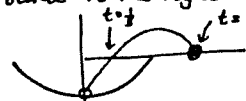
$$\begin{aligned} \text{b) (i) } \ddot{x} &= 0 \\ \ddot{x} &= c_1 \\ \text{When } t=0 \quad \dot{x} &= V \cos \alpha \\ &= 12 \cos \alpha \\ \therefore c_1 &= 12 \cos \alpha \\ \dot{x} &= 12 \cos \alpha \\ x &= 12t \cos \alpha + c_2 \\ \text{When } t=0 \quad x &= 0 \therefore c_2 = 0 \\ x &= (12 \cos \alpha)t \end{aligned}$$

$$\begin{aligned} \ddot{y} &= -10 \\ y &= -10t^2 + c_3 \\ \text{When } t=0 \quad \dot{y} &= V \sin \alpha \\ &= 12 \sin \alpha \\ \therefore c_3 &= 12 \sin \alpha \\ y &= -5t^2 + 12t \sin \alpha + c_4 \\ \text{When } t=0 \quad y &= -1 \text{ (starts at bottom of bunker).} \\ \therefore c_4 &= -1 \\ y &= -5t^2 + (2 \sin \alpha)t - 1 \end{aligned}$$

$$\begin{aligned} \text{(ii) } \alpha &= 30^\circ \text{ Ball will hit ground when } y=0 \\ y &= -5t^2 + (2 \sin \alpha)t - 1 \\ 0 &= -5t^2 + (2 \sin 30^\circ)t - 1 \\ 0 &= -5t^2 + 6t - 1 \\ 5t^2 - 6t + 1 &= 0 \\ (5t-1)(t-1) &= 0 \\ t &= \frac{1}{5} \text{ or } t=1 \end{aligned}$$

$t = \frac{1}{5}$ gives first time ball crosses x -axis which is not on the ground (it is left of A).

$t=1$ gives the time the ball hits ground to the right of A



$$\begin{aligned} \text{When } t=1 \quad x &= (12 \cos 30^\circ) \times 1 \\ &= 12 \times \frac{\sqrt{3}}{2} \\ &= 6\sqrt{3} \end{aligned}$$

But OA = 4 metres, so ball will land 1.6 metres from A $6\sqrt{3}-4$

(ii) For the ball to land to the right of A look at angle necessary to go through A
A is $(4, 0)$

$$\begin{aligned} 4 &= (2 \cos \alpha)t \quad \text{sub into } y = -5t^2 + (2 \sin \alpha)t - 1 \\ t &= \frac{4}{2 \cos \alpha} = \frac{2}{\cos \alpha} \quad \text{when } y=0 \end{aligned}$$

$$0 = -5 \left(\frac{1}{\cos^2 \alpha} \right) + (2 \sin \alpha) \left(\frac{1}{\cos \alpha} \right) - 1$$

$$0 = -\frac{5}{\cos^2 \alpha} + \frac{2 \sin \alpha}{\cos \alpha} - 1$$

$$\begin{aligned} 0 &= -5(1 + \tan^2 \alpha) + 36 \tan \alpha - 9 \quad (\times 9) \\ &= -5 - 5 \tan^2 \alpha + 36 \tan \alpha - 9 \end{aligned}$$

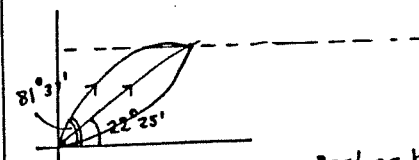
$$5 \tan^2 \alpha - 36 \tan \alpha + 14 = 0$$

Use formula

$$\tan \alpha = \frac{36 \pm \sqrt{36^2 - 4 \times 5 \times 14}}{10}$$

$$= 0.4125245 \text{ or } 6.78$$

$$\alpha = 22^\circ 25' \text{ or } 81^\circ 37' \text{ (first quadrant only)}$$



Anything less than $22^\circ 25'$ or bigger than $81^\circ 37'$ will hit the bank of the bunker

So, to land to the right of A

$$23^\circ \leq \alpha \leq 81^\circ \text{ (to nearest degree)}$$